

# Full Stack / ServiceNow Flow Designer

## Project Documentation Format

|              |                                                |               |                          |
|--------------|------------------------------------------------|---------------|--------------------------|
| Date         | 03/10/2026                                     | Team ID       | 6ab8f707c23107aaaa73276f |
| Project Name | Auto Ticket Classification using Flow Designer | Maximum Marks | 10 Marks                 |

### 1. Introduction

- Project Title: Auto Ticket Classification using Flow Designer
- Team ID: 6ab8f707c23107aaaa73276f
- Team Members & Roles: Lead Developer (Flow Designer Architect), ServiceNow Administrator (Schema & Security), QA & Automation Tester (UAT & Performance), Process Analyst (ITIL Documentation).

### 2. Project Overview

- Purpose: Automate the categorization, subcategorization, and notification of IT support incidents in ServiceNow without custom JavaScript or manual triage.
- Features: Real-time record trigger on incident insert, conditional keyword parsing on Short Description, automated Category/Subcategory assignment, automated transactional email notification to caller, and unclassified incident fallback handling.

### 3. Architecture

- Frontend / Client Layer: ServiceNow Service Portal & Native Incident Form UI (HTML5/AngularJS ServiceNow client framework).
- Workflow & Execution Layer: ServiceNow Flow Designer Engine operating on an asynchronous event queue.
- Database / Storage Layer: ServiceNow CMDB / Relational Cloud Database, primarily targeting the Incident [incident] table with foreign keys referencing User [sys\_user] and Choice [sys\_choice] tables.

### 4. Setup Instructions

- Prerequisites: ServiceNow Personal Developer Instance (PDI) with Washington DC or Vancouver release, System Administrator role (admin).
- Installation: 1. Navigate to System Update Sets → Retrieved Update Sets.  
2. Import XML file: 'AutoTicketClassification\_v1.0.xml'.  
3. Preview and Commit Update Set.  
4. Open Flow Designer and verify the flow 'Auto Ticket Classification' is set to Active.

### 5. Folder & Component Structure

- Flow Components:
  - Trigger: Record Created (Table: incident, Condition: Category IS EMPTY)
  - Decision Nodes: Flow Logic If / Else If (Keyword checks on short\_description)
  - Action Steps: Update Record (Set Category, Subcategory), Send Email (sys\_user caller notification)
- System Definitions: Data Dictionary overrides on incident choice fields; outbound email template.

### 6. Running the Application

1. Navigate to Incident → Create New in ServiceNow application navigator.
2. Select a Caller and enter an issue in the Short Description (e.g., 'WiFi not working in library').
3. Leave Category and Subcategory blank.

4. Click Submit.
5. Reopen the ticket to verify automated assignment (Category: Network, Subcategory: Wi-Fi).
6. Check System Logs → Emails to confirm automated notification.

## 7. API & Trigger Documentation

- Trigger Endpoint: Internal GlideRecord Insert Event on Table [incident]
- Filter Query: `active=true^category!EMPTY`
- Action 1: `sn\_fd.FlowAPI.executeFlow('global.auto\_ticket\_classification', inputs)`
- Action 2 (Email Dispatch): Target `sys\_email`, Type: `send-ready`, Recipient: `incident.caller\_id.email`, Subject: `Incident \${number} Classified & Received`.

## 8. Authentication & Security

- Access Control: Handled via ServiceNow Role-Based Access Control (RBAC).
- Roles Enforced: `itil` role required to view and modify incident records; `admin` role required to edit Flow Designer definitions.
- Flow Context: Configured to execute as 'System User' to ensure uninterrupted execution regardless of portal user ACL restrictions.

## 9. User Interface

- Standard Incident Form View (`incident.do`) displaying auto-populated dependent dropdowns.
- Service Portal Ticket Status Page displaying real-time updates and categorization to end-users.
- Flow Execution Execution Details view providing step-by-step runtime variable state visualization.

## 10. Testing Strategy

- Strategy: Automated Test Framework (ATF) combined with manual User Acceptance Testing (UAT).
- Scope: Positive keyword classification, case-insensitivity tests, guardrail condition preservation, notification dispatch timing, and high-concurrency batch submission.

## 11. Screenshots or Demo Link

- Live Demo Sequence: Creation of ticket → Automatic field population → Flow Designer Execution Context → Received Email Notification.
- Repository / Demo Artifact: ServiceNow Developer Instance Git Repository Link & Update Set XML package.

## 12. Known Issues & Limitations

- Multi-Keyword Ambiguity: Tickets containing multiple contrasting keywords (e.g., 'Projector screen WiFi broken') match the first sequential flow condition.
- Unlisted Vocabularies: Slang or obscure abbreviations not in the rule matrix route to the default triage fallback queue.

## 13. Future Enhancements

- Integration with ServiceNow Predictive Intelligence (ML Classification Model) to handle freeform semantic descriptions.
- Automated Assignment Group mapping based on technician skills and geographical location.
- Interactive Virtual Agent (Chatbot) integration for real-time deflection.